

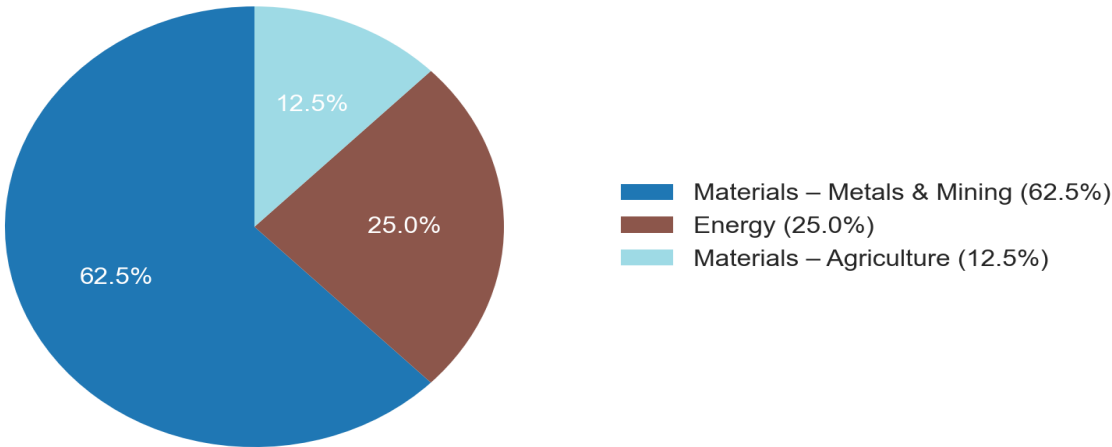
# Portfolio rotation report

|                    |                                                                                                                                                                                                                                                                                                                                                             |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Date of extraction | 2025-12-01 17:48                                                                                                                                                                                                                                                                                                                                            |
| Data window        | 2018-01-01 → 2025-10-20                                                                                                                                                                                                                                                                                                                                     |
| Frequency          | <b>Weekly</b> — 408 observations.<br><b>Sufficient depth:</b> short- and long-term indicators are statistically reliable.<br><b>Warning:</b> the observation window exceeds the typical interpretability range for this frequency (+684.6% above the recommended upper bound). Results may blend multiple market regimes, reducing economic signal clarity. |
| Tickers            | DBA, DBC, GLD, GSG, IAU, SLV, USO, XLE                                                                                                                                                                                                                                                                                                                      |
| Indicators         | <b>Short term:</b> 8 weeks volatility, 12 weeks return, 16 weeks momentum<br><b>Long term:</b> 28 weeks volatility, 30 weeks return, 80 weeks momentum                                                                                                                                                                                                      |

## 1. Portfolio Composition

- XLE (Energy Select Sector SPDR Fund — Energy)
- GLD (SPDR Gold Shares — Materials – Metals & Mining)
- SLV (iShares Silver Trust — Materials – Metals & Mining)
- DBC (Invesco DB Commodity Index Tracking Fund — Materials – Metals & Mining)
- USO (United States Oil Fund — Energy)
- DBA (Invesco DB Agriculture Fund — Materials – Agriculture)
- GSG (iShares S&P; GSCI Commodity-Indexed Trust — Materials – Metals & Mining)
- IAU (iShares Gold Trust — Materials – Metals & Mining)

Sector Allocation



## Macro Profile Insight

### Macro coverage

- Highly concentrated structure: limited diversification and sensitivity to one macro driver.

### Structural properties

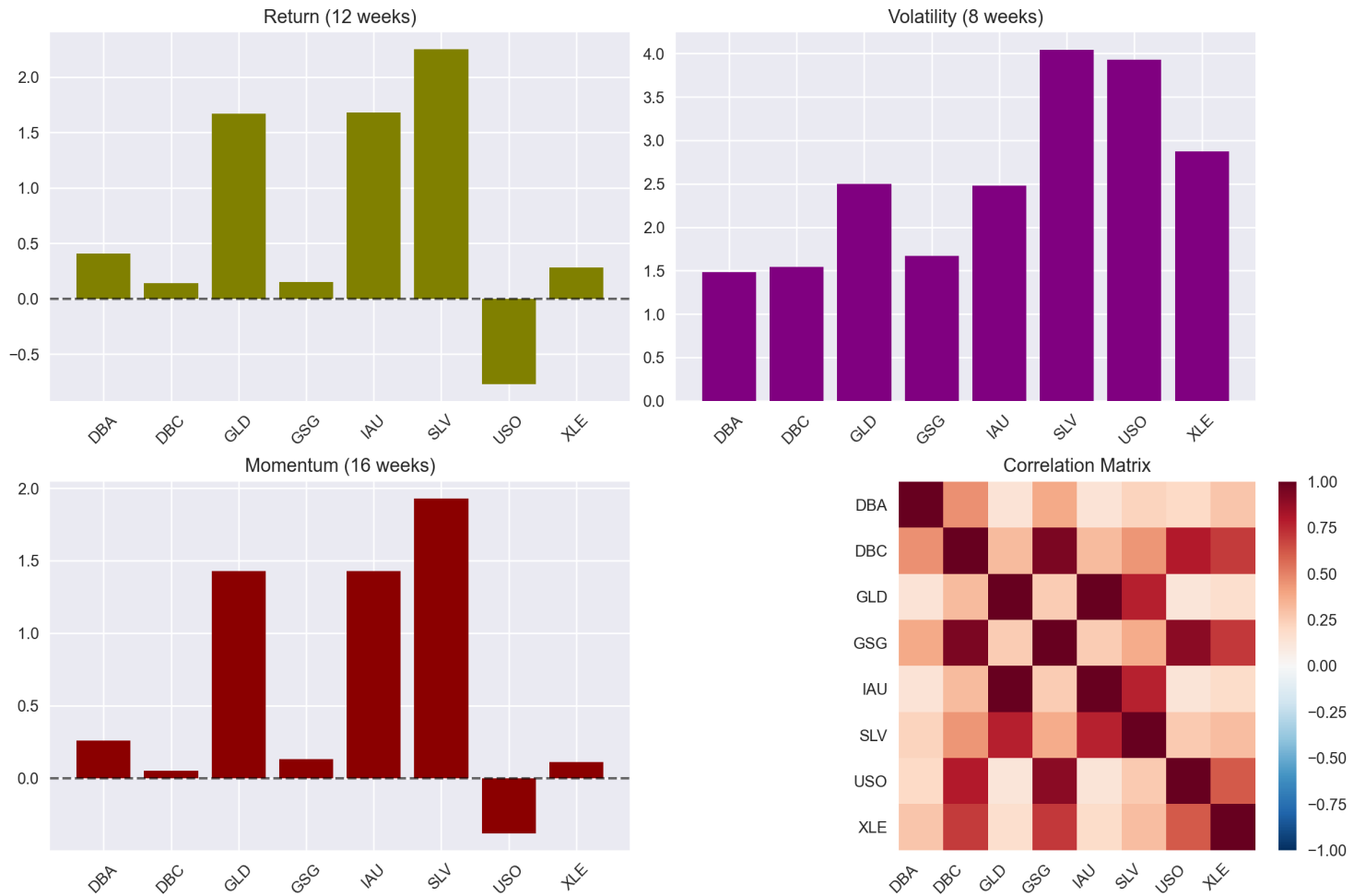
- Large/mega-cap tilt: quality and stability prioritized over idiosyncratic risk.
- Concentration risk: portfolio heavily exposed to etf instruments.
- Liquidity stance: pockets of thin/illiquid names.
- Defensive beta tilt: lower systematic risk profile.
- Balanced volatility structure: risk evenly distributed across holdings.

### Additional comments

- Significant sector gaps: broad portions of the economy remain uncovered.

2. Sector Performance Visualization

Sector Performance Dashboard



Top Positive Correlations  
71.4% of portfolio

| Pair      | Correlation |
|-----------|-------------|
| GLD / IAU | 1.0         |
| DBC / GSG | 0.94        |
| GSG / USO | 0.9         |
| DBC / USO | 0.79        |
| GLD / SLV | 0.78        |

Top Neutral Correlations  
28.6% of portfolio

| Pair      | Correlation |
|-----------|-------------|
| GLD / USO | 0.12        |
| IAU / USO | 0.13        |
| DBA / GLD | 0.14        |
| DBA / IAU | 0.14        |
| GLD / XLE | 0.17        |

Overall, average cross-sector correlation was 0.41, ranging from 0.12 to 0.94. This suggests elevated co-movement — diversification benefits are partially reduced. Correlation dispersion is wide, showing sharp contrasts between sectors. Strongest co-movement between GLD and IAU (1.00); weakest between GLD and USO (0.12). Mixed regime: both market-wide and sector-specific forces are at play.

### 3. Key Performance Indicators

#### 3.a Current Regime Snapshot

| Ticker | R (%) | V (%) | RAR    | M (%) | Signal            | Regime   | Liquidity | Commentary                                                                                                                                                         |
|--------|-------|-------|--------|-------|-------------------|----------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DBA    | 0.41  | 1.48  | 0.277  | 0.26  | neutral           | normal   | illiquid  | mixed short-term dynamics, signal unclear or in transition, normal volatility regime, under limited liquidity.                                                     |
| DBC    | 0.14  | 1.54  | 0.091  | 0.05  | neutral           | normal   | illiquid  | mixed short-term dynamics, signal unclear or in transition, normal volatility regime, under limited liquidity.                                                     |
| GLD    | 1.67  | 2.50  | 0.668  | 1.43  | mildly coherent   | elevated | thin      | positive performance with solid momentum, signal remains coherent, but with limited conviction, rising volatility — tension building, under limited liquidity.     |
| GSG    | 0.15  | 1.67  | 0.09   | 0.13  | neutral           | normal   | thin      | mixed short-term dynamics, signal unclear or in transition, normal volatility regime, under limited liquidity.                                                     |
| IAU    | 1.68  | 2.48  | 0.677  | 1.43  | mildly coherent   | elevated | thin      | positive performance with solid momentum, signal remains coherent, but with limited conviction, rising volatility — tension building, under limited liquidity.     |
| SLV    | 2.25  | 4.04  | 0.557  | 1.93  | strongly coherent | elevated | thin      | positive performance with solid momentum, signal remains strongly coherent with long-term momentum, rising volatility — tension building, under limited liquidity. |
| USO    | -0.77 | 3.93  | -0.196 | -0.38 | neutral           | normal   | thin      | mixed short-term dynamics, signal unclear or in transition, normal volatility regime, under limited liquidity.                                                     |
| XLE    | 0.28  | 2.87  | 0.098  | 0.11  | neutral           | subdued  | thin      | mixed short-term dynamics, signal unclear or in transition, low volatility — calm market tone, under limited liquidity.                                            |

#### Top by Average Return

| Ticker | AvgReturn(%) |
|--------|--------------|
| SLV    | 2.25         |
| IAU    | 1.68         |
| GLD    | 1.67         |
| DBA    | 0.41         |
| XLE    | 0.28         |

#### Top by Volatility

| Ticker | Volatility(%) |
|--------|---------------|
| SLV    | 4.04          |
| USO    | 3.93          |
| XLE    | 2.87          |
| GLD    | 2.5           |
| IAU    | 2.48          |

#### Top by Momentum

| Ticker | Momentum(%) |
|--------|-------------|
| SLV    | 1.93        |
| GLD    | 1.43        |
| IAU    | 1.43        |
| DBA    | 0.26        |
| GSG    | 0.13        |

#### Last by Average Return

| Ticker | AvgReturn(%) |
|--------|--------------|
| USO    | -0.77        |
| DBC    | 0.14         |
| GSG    | 0.15         |
| XLE    | 0.28         |
| DBA    | 0.41         |

#### Last by Volatility

| Ticker | Volatility(%) |
|--------|---------------|
| DBA    | 1.48          |
| DBC    | 1.54          |
| GSG    | 1.67          |
| IAU    | 2.48          |
| GLD    | 2.5           |

#### Last by Momentum

| Ticker | Momentum(%) |
|--------|-------------|
| USO    | -0.38       |
| DBC    | 0.05        |
| XLE    | 0.11        |
| GSG    | 0.13        |
| DBA    | 0.26        |

### 3.b Rotation & Stress Metrics

| Ticker | ΔReturn (pp) | Momentum Ratio | Signal | Volatility Ratio | Beta | ADV10 USD | Interpretation                                                                                                                     |
|--------|--------------|----------------|--------|------------------|------|-----------|------------------------------------------------------------------------------------------------------------------------------------|
| DBA    | 0.31         | 2.01           | 0.086  | 0.86             | 0.24 | 239.2 K   | stable performance, momentum surging, volatility normalizing, within defensive profile, under thin liquidity regime.               |
| DBC    | 0.04         | -0.50          | -0.033 | 0.70             | 0.28 | 523.4 K   | stable performance, momentum collapsing, volatility compressing, within defensive profile, under thin liquidity regime.            |
| GLD    | 0.76         | 1.55           | 0.561  | 0.99             | 0.05 | 10.79 M   | acceleration in returns, momentum surging, volatility normalizing, within defensive profile, within normal liquidity conditions.   |
| GSG    | 0.05         | 1.30           | 0.057  | 0.75             | 0.34 | 1.02 M    | stable performance, momentum strengthening, volatility compressing, within defensive profile, under thin liquidity regime.         |
| IAU    | 0.76         | 1.54           | 0.564  | 0.99             | 0.04 | 10.01 M   | acceleration in returns, momentum surging, volatility normalizing, within defensive profile, within normal liquidity conditions.   |
| SLV    | 1.18         | 2.17           | 0.609  | 1.25             | 0.39 | 27.50 M   | acceleration in returns, momentum surging, volatility expanding, within defensive profile, within normal liquidity conditions.     |
| USO    | -0.46        | 2.27           | 0.116  | 0.82             | 0.46 | 4.76 M    | stable performance, momentum surging, volatility normalizing, within defensive profile, under thin liquidity regime.               |
| XLE    | 0.45         | -1.10          | -0.052 | 1.06             | 0.80 | 16.74 M   | stable performance, momentum collapsing, volatility normalizing, aligned with market dynamics, within normal liquidity conditions. |

#### ■ Moderate-impact observations

For **DBA, DBC, GSG, XLE**: *Technical rebound* — improving momentum while returns are still limited. (avg ann. return 0.2%, momentum 0.1%, vol 1.9%)

For **SLV**: *Rally under tension* — strong gains while volatility remains high — vulnerable to pullbacks. (avg ann. return 2.2%, momentum 1.9%, vol 4.0%)

#### ■ Light-impact observations

For **GLD, IAU, USO**: *Range / noise* — no clear direction; rotating momentum and mid-range volatility. (avg ann. return 0.9%, momentum 0.8%, vol 3.0%)